Confidence Interval: unknown sigma

We want to calculate a confidence interval but since the population standard deviation is almost always unknown what do we do? Instead of using the normal distribution we switch to Student's t-distribution. This changes the margin of error calculation, but the rest of the procedure remains the same.

We need a few criteria to be satisfied:

- A simple random sample is used.
- The population standard deviation, σ, is unknown.
- The population distribution is normal or a sample size of $n \geq 30$ is used.

In this case the margin of error is given by:

$$\text{MOE} = t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation and n is the sample size.

So the only remaining step is to determine the critical value: $t_{\frac{\alpha}{2}}$. As before, we find α depending on the level on confidence chosen. But to use the t-distribution table we also need to know the degree of freedom. It turns out that this is 1 less than the sample size: $\text{df} = n - 1$.

We will use the same distribution as we used for the known σ notes: a normal distribution with μ = 68 and standard deviation of σ = 3: N(68, 3).

```
μ = 68
```

```
σ = 3
```

We generating 100,000 points from this distribution as shown below:

```
height_data =
  [66.1135, 67.551, 66.3548, 71.5904, 67.181, 61.4269, 71.8204, 69.2633, 72.6117, 65.2339, 68.79
```

Confidence Level: 80%, Sample size: 5

We are going to compare the confidence interval for a known sigma versus unknown sigma.

$$\alpha = 0.2, \quad \alpha/2 = 0.1, \quad t_{\alpha/2} = 1.533 \quad z_{\alpha/2} = 1.28$$

Let's look at what changing the sample size does to a confidence interval:

Confidence Level: 80%

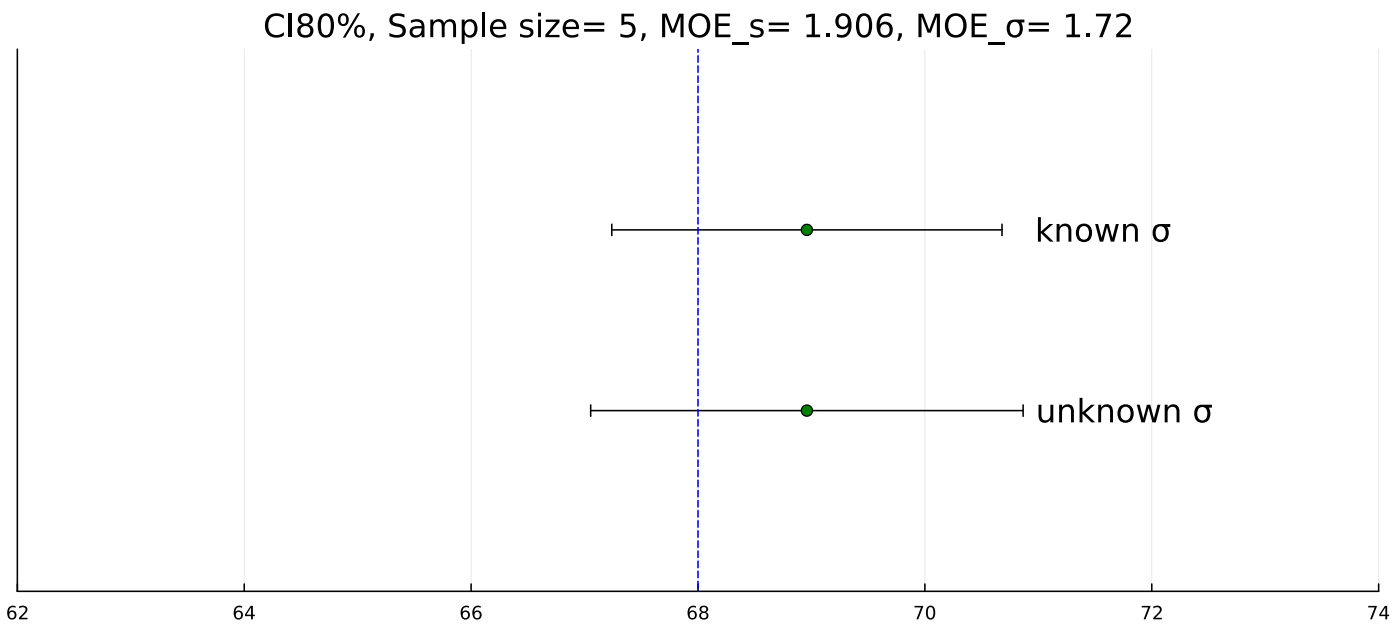
Gather another sample

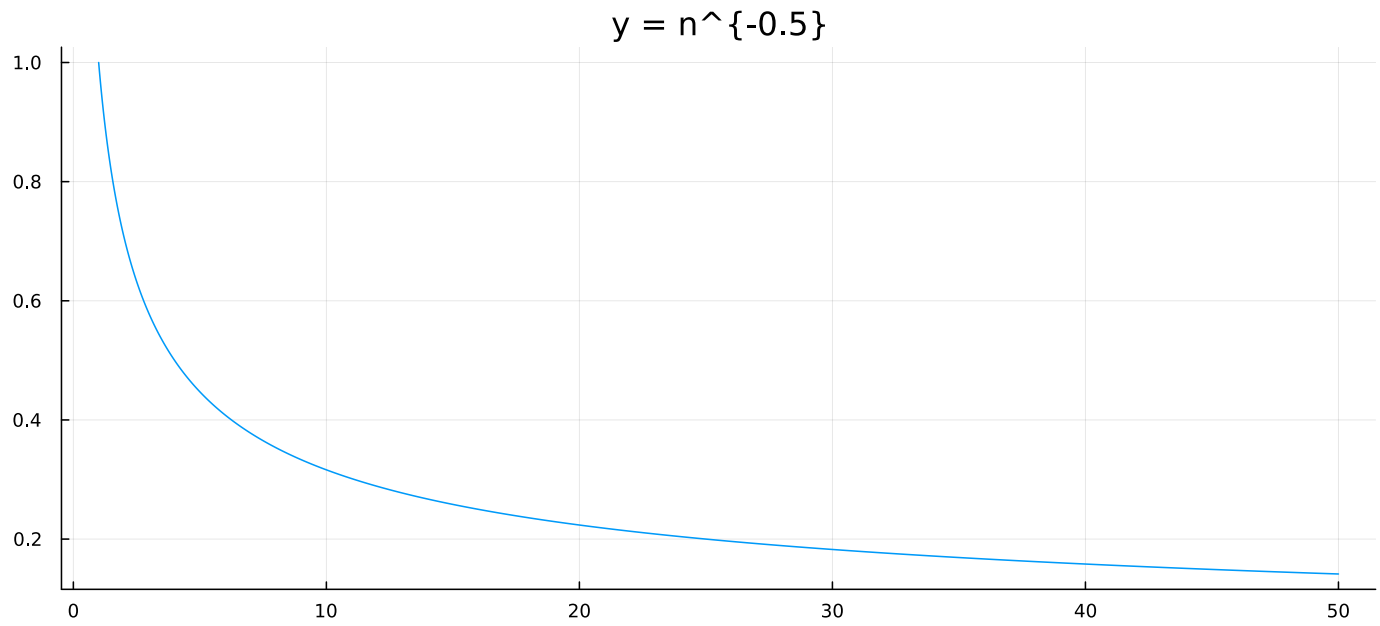
Sample data: 66.41, 69.69, 67.89, 73.46, 67.36

$$\bar{x} = 68.96, \quad s = 2.78, \quad \sigma = 3$$

$$\text{MOE}_s = t_{\alpha/2}\frac{s}{\sqrt{5}} = 1.533\frac{2.78}{\sqrt{5}} = 1.906, \quad \text{MOE}_\sigma = z_{\alpha/2}\frac{\sigma}{\sqrt{5}} = 1.28\frac{3}{\sqrt{5}} = 1.72$$

Confidence Level: 80%



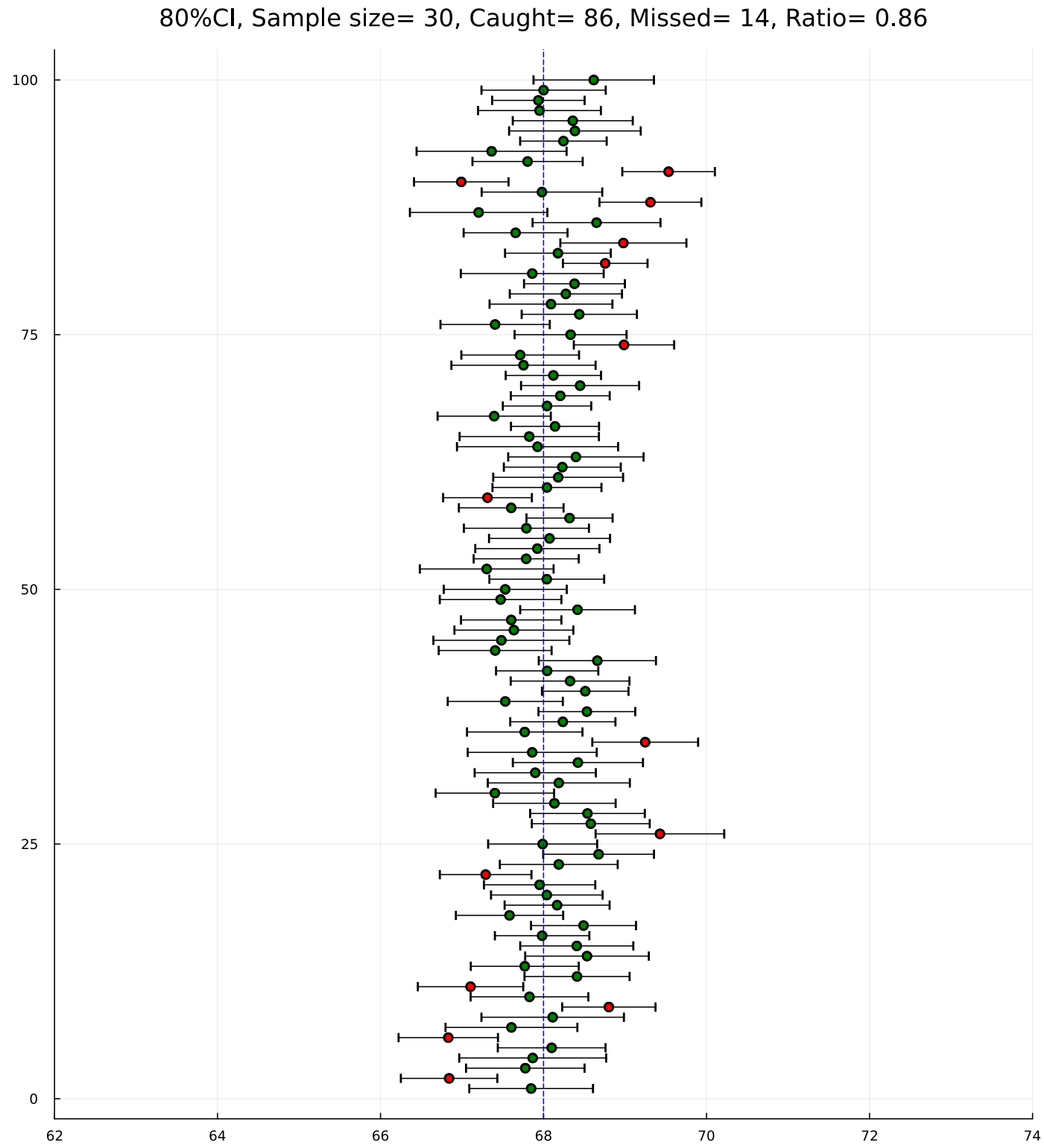


Number of CIs 100, Sample size: 30

$$\alpha = 0.2, \quad \text{Critical value: } t_{\alpha/2} = 1.311$$

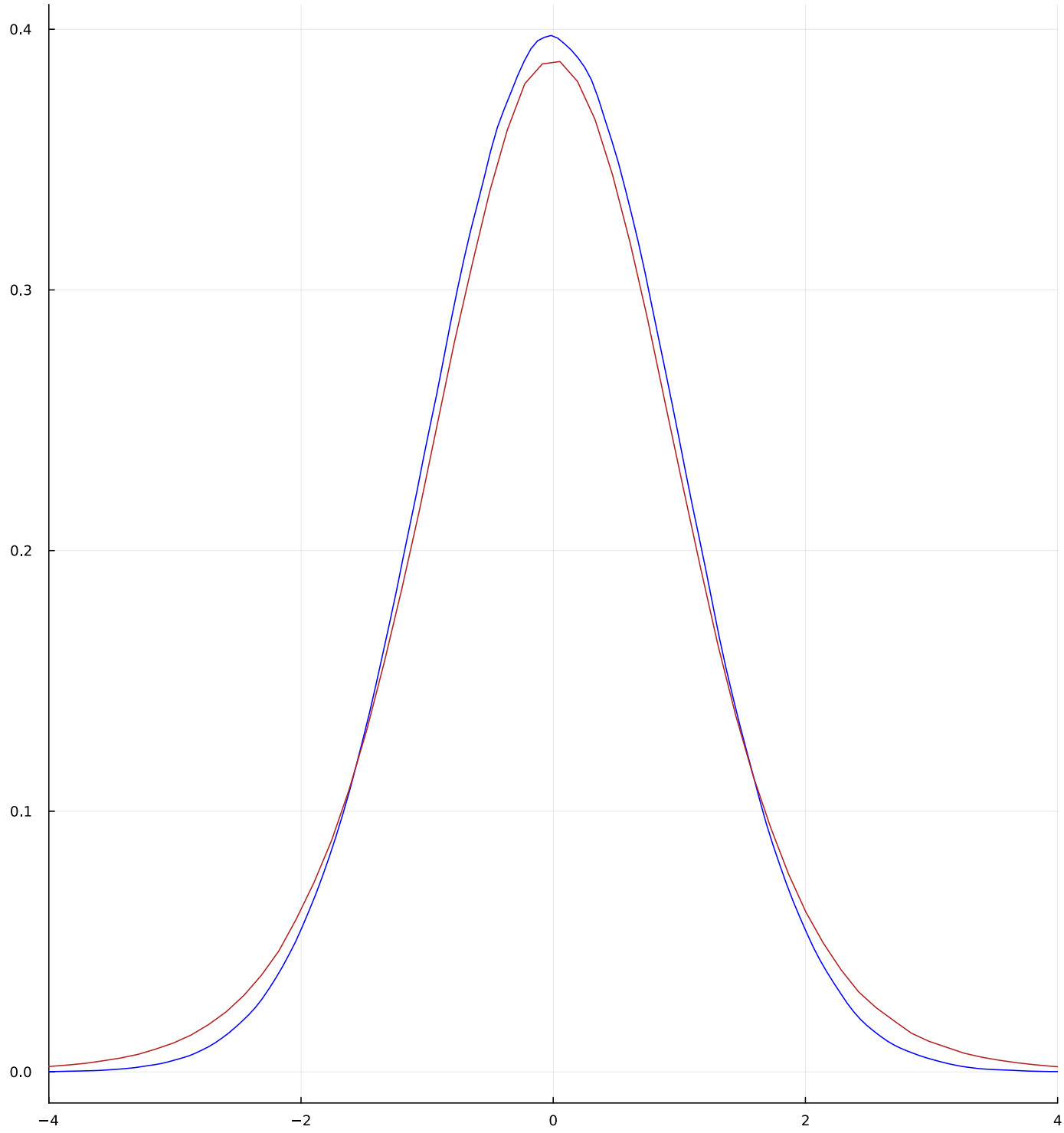
Hiding the code to get sample means here 👀

Here be the hidden code to set colors and calculate caught ratio and possibly a 🐍



Normal vs t-distribution graph comparison needs work.

df = 10




```
1 Enter cell code...
```